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Determine the value of $\frac{D}{T}$ as a percentage.

1. This indicates that 50% of the shoppers bought Brand X in the first month, so $A = 0.5T$. It follows that the other 50% of the shoppers bought Brand Y in the first month, so $B = 0.5T$. Then, $D = 0.1A + 0.6B = 0.1(0.5T) + 0.6(0.5T) = 0.05T + 0.30T = 0.35T$. It follows that $\frac{D}{T} = \frac{0.35T}{T} = 0.35$, which is 35%; SUFFICIENT.
2. This indicates that $T = 5{,}000$, as shown in the following table:

		Second Month		
		X	**Y**	**Total**
First Month	**X**	$0.9A$	$0.1A$	A
	Y	$0.4B$	$0.6B$	B
	Total	C	D	5,000

But not enough information is given to be able to determine D or D as a percentage of 5,000; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

•

If the total price of n equally priced shares of a certain stock was $12,000, what was the price per share of the stock?

1. If the price per share of the stock had been $1 more, the total price of the n shares would have been $300 more.
2. If the price per share of the stock had been $2 less, the total price of the n shares would have been 5 percent less.

Arithmetic Arithmetic Operations; Percents

Since the price per share of the stock can be expressed as $\frac{\$12{,}000}{n}$, determining the value of n is sufficient to answer this question.

1. A per-share increase of $1 and a total increase of $300 for n shares of stock mean together that $n(\$1) = \$300$. It follows that $n = 300$; SUFFICIENT.
2. If the price of each of the n shares had been reduced by $2, the total reduction in price would have been 5 percent less or 0.05($12,000). The equation $2n = 0.05(\$12{,}000)$ expresses this relationship. The value of n can be determined to be 300 from this equation; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

•

In Year X, 8.7 percent of the men in the labor force were unemployed in June compared with 8.4 percent in May. If the number of men in the labor force was the same for both months, how many men were unemployed in June of that year?

1. In May of Year X, the number of unemployed men in the labor force was 3.36 million.
2. In Year X, 120,000 more men in the labor force were unemployed in June than in May.

Arithmetic Percents

Since 8.7 percent of the men in the labor force were unemployed in June, the number of unemployed men could be calculated if the total number of men in the labor force was known. Let t represent the total number of men in the labor force.

1. This implies that for May $(8.4\%)t = 3{,}360{,}000$, from which the value of t can be determined; SUFFICIENT.

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1. This implies that for May $(8.4\%)t = 3{,}360{,}000$, from which the value of t can be determined; SUFFICIENT.
2. This implies that $(8.7\% - 8.4\%)t = 120{,}000$ or $(0.3\%)t = 120{,}000$. This equation can be solved for t; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

•

On Monday morning a certain machine ran continuously at a uniform rate to fill a production order. At what time did it completely fill the order that morning?

1. The machine began filling the order at 9:30 a.m.
2. The machine had filled $\frac{1}{2}$ of the order by 10:30 a.m. and $\frac{5}{6}$ of the order by 11:10 a.m.

Arithmetic Arithmetic Operations

1. This merely states what time the machine began filling the order; NOT sufficient.
2. In the 40 minutes between 10:30 a.m. and 11:10 a.m., $\frac{5}{6} - \frac{1}{2} = \frac{1}{3}$ of the order was filled. Therefore, the entire order was completely filled in 3 × 40 = 120 minutes, or 2 hours. Since half the order took 1 hour and was filled by 10:30 a.m., the second half of the order, and thus the entire order, was filled by 11:30 a.m.; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

•

After winning 50 percent of the first 20 games it played, Team A won all of the remaining games it played. What was the total number of games that Team A won?

1. Team A played 25 games altogether.
2. Team A won 60 percent of all the games it played.

Arithmetic Percents

Let r be the number of the remaining games played, all of which the team won. Since the team won (50%)(20) = 10 of the first 20 games and the r remaining games, the total number of games the team won is $10 + r$. Also, the total number of games the team played is $20 + r$. Determine the value of r.

1. Given that the total number of games played is 25, it follows that $20 + r = 25$, or $r = 5$; SUFFICIENT.
2. It is given that the total number of games won is $(60\%)(20 + r)$, which can be expanded as $12 + 0.6r$. Since it is also known that the number of games won is $10 + r$, it follows that $12 + 0.6r = 10 + r$. Solving this equation gives $12 - 10 = r - 0.6r$, or $2 = 0.4r$, or $r = 5$; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

•

Michael arranged all his books in a bookcase with 10 books on each shelf and no books left over. After Michael acquired 10 additional books, he arranged all his books in a new bookcase with 12 books on each shelf and no books left over. How many books did Michael have before he acquired the 10 additional books?

1. Before Michael acquired the 10 additional books, he had fewer than 96 books.
2. Before Michael acquired the 10 additional books, he had more than 24 books.

Arithmetic Properties of Numbers

If x is the number of books Michael had before he acquired the 10 additional books, then x is a multiple of 10. After Michael acquired the 10 additional books, he had $x + 10$ books and $x + 10$ is a multiple of 12.

1. If $x < 96$, where x is a multiple of 10, then $x = 10, 20, 30, 40, 50, 60, 70, 80$, or 90 and $x + 10 = 20, 30, 40, 50, 60, 70, 80, 90$, or 100. Since $x + 10$ is a multiple of 12, then $x + 10 = 60$ and $x = 50$; SUFFICIENT.
2. If $x > 24$, where x is a multiple of 10, then x must be one of the numbers 30, 40, 50, 60, 70, 80, 90, 100, 110, …, and $x + 10$ must be one of the numbers 40, 50, 60, 70, 80, 90, 100, 110, 120, …. Since there is more than

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one multiple of 12 among these numbers (for example, 60 and 120), the value of $x + 10$, and therefore the value of x, cannot be determined; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

•

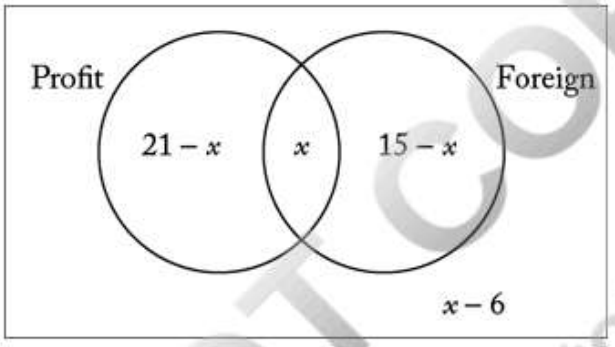
Last year in a group of 30 businesses, 21 reported a net profit and 15 had investments in foreign markets. How many of the businesses did not report a net profit nor invest in foreign markets last year?

1. Last year 12 of the 30 businesses reported a net profit and had investments in foreign markets.
2. Last year 24 of the 30 businesses reported a net profit or invested in foreign markets, or both.

Arithmetic Concepts of Sets

Consider the Venn diagram below in which x represents the number of businesses that reported a net profit and had investments in foreign markets. Since 21 businesses reported a net profit, $21 - x$ businesses reported a net profit only. Since 15 businesses had investments in foreign markets, $15 - x$ businesses had investments in foreign markets only. Finally, since there is a total of 30 businesses, the number of businesses that did not report a net profit and did not invest in foreign markets is $30 - (21 - x + x + 15 - x) = x - 6$.

Determine the value of $x - 6$, or equivalently, the value of x.



1. It is given that $12 = x$; SUFFICIENT.
2. It is given that $24 = (21 - x) + x + (15 - x)$. Therefore, $24 = 36 - x$, or $x = 12$.

   Alternatively, the information given is exactly the number of businesses that are not among those to be counted in answering the question posed in the problem, and therefore the number of businesses that are to be counted is 30 − 24 = 6; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

•

For each landscaping job that takes more than 4 hours, a certain contractor charges a total of r dollars for the first 4 hours plus $0.2r$ dollars for each additional hour or fraction of an hour, where $r > 100$. Did a particular landscaping job take more than 10 hours?

1. The contractor charged a total of $288 for the job.
2. The contractor charged a total of $2.4r$ dollars for the job.

Algebra Applied Problems

If y represents the total number of hours the particular landscaping job took, determine if $y > 10$.

1. This indicates that the total charge for the job was $288, which means that $r + 0.2r(y - 4) = 288$. From this it cannot be determined if $y > 10$. For example, if $r = 120$ and $y = 11$, then 120 + 0.2(120)(7) = 288, and the job took more than 10 hours. However, if $r = 160$ and $y = 8$, then 160 + 0.2(160)(4) = 288, and the job took less than 10 hours; NOT sufficient.
2. This indicates that $r + 0.2r(y - 4) = 2.4r$, from which it follows that

$$r + 0.2ry - 0.8r = 2.4r \quad \text{use distributive property}$$

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$0.2ry = 2.2r$ subtract $(r - 0.8r)$ from both sides

$y = 11$ divide both sides by $0.2r$

Therefore, the job took more than 10 hours; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

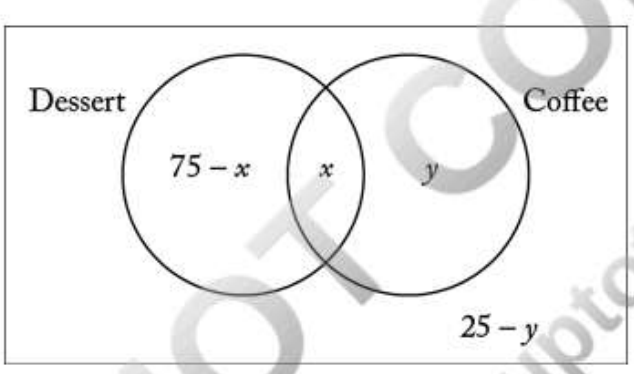
•

If 75 percent of the guests at a certain banquet ordered dessert, what percent of the guests ordered coffee?

1. 60 percent of the guests who ordered dessert also ordered coffee.
2. 90 percent of the guests who ordered coffee also ordered dessert.

Arithmetic Concepts of Sets; Percents

Consider the Venn diagram below that displays the various percentages of 4 groups of the guests. Thus, x percent of the guests ordered both dessert and coffee and y percent of the guests ordered coffee only. Since 75 percent of the guests ordered dessert, $(75 - x)\%$ of the guests ordered dessert only. Also, because the 4 percentages represented in the Venn diagram have a total sum of 100 percent, the percentage of guests who did not order either dessert or coffee is $100 - [(75 - x) + x + y] = 25 - y$. Determine the percentage of guests who ordered coffee, or equivalently, the value of $x + y$.



1. Given that x is equal to 60 percent of 75, or 45, the value of $x + y$ cannot be determined; NOT sufficient.
2. Given that 90 percent of $x + y$ is equal to x, it follows that $0.9(x + y) = x$, or $9(x + y) = 10x$. Therefore, $9x + 9y = 10x$, or $9y = x$. From this the value of $x + y$ cannot be determined. For example, if $x = 9$ and $y = 1$, then all 4 percentages in the Venn diagram are between 0 and 100, $9y = x$, and $x + y = 10$. However, if $x = 18$ and $y = 2$, then all 4 percentages in the Venn diagram are between 0 and 100, $9y = x$, and $x + y = 20$; NOT sufficient.

Given both (1) and (2), it follows that $x = 45$ and $9y = x$. Therefore, $9y = 45$, or $y = 5$, and hence $x + y = 45 + 5 = 50$.

The correct answer is C; both statements together are sufficient.

•

A tank containing water started to leak. Did the tank contain more than 30 gallons of water when it started to leak? (Note: 1 gallon = 128 ounces)

1. The water leaked from the tank at a constant rate of 6.4 ounces per minute.
2. The tank became empty less than 12 hours after it started to leak.

Arithmetic Rate Problems

1. Given that the water leaked from the tank at a constant rate of 6.4 ounces per minute, it is not possible to determine if the tank leaked more than 30 gallons of water. In fact, any nonzero amount of water leaking from the tank is consistent with a leakage rate of 6.4 ounces per minute, since nothing can be determined about the amount of time the water was leaking from the tank; NOT sufficient.
2. Given that the tank became empty in less than 12 hours, it is not possible to determine if the tank leaked more than 30 gallons of water because the rate at which water leaked from the tank is unknown. For example, the tank

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than 30 gallons of water because the rate at which water leaked from the tank is unknown. For example, the tank could have originally contained 1 gallon of water that emptied in exactly 10 hours or the tank could have originally contained 31 gallons of water that emptied in exactly 10 hours; NOT sufficient.

Given (1) and (2) together, the tank emptied at a constant rate of $\left(6.4\frac{\text{oz}}{\text{min}}\right)\left(60\frac{\text{min}}{\text{hr}}\right)\left(\frac{1}{128}\frac{\text{gal}}{\text{oz}}\right) = \frac{(64)(6)}{128}\frac{\text{gal}}{\text{hr}} = \frac{(64)(6)}{(64)(2)}\frac{\text{gal}}{\text{hr}} = 3\frac{\text{gal}}{\text{hr}}$ for less than 12 hours. If t is the total number of hours the water leaked from the tank, then the total amount of water emptied from the tank, in gallons, is $3t$, which is therefore less than (3)(12) = 36. From this it is not possible to determine if the tank originally contained more than 30 gallons of water. For example, if the tank leaked water for a total of 11 hours, then the tank originally contained (3)(11) gallons of water, which is more than 30 gallons of water. However, if the tank leaked water for a total of 2 hours, then the tank originally contained (3)(2) gallons of water, which is not more than 30 gallons of water.

The correct answer is E; both statements together are still not sufficient.

•

Each of the 45 books on a shelf is written either in English or in Spanish, and each of the books is either a hardcover book or a paperback. If a book is to be selected at random from the books on the shelf, is the probability less than $\frac{1}{2}$ that the book selected will be a paperback written in Spanish?

1. Of the books on the shelf, 30 are paperbacks.
2. Of the books on the shelf, 15 are written in Spanish.

Arithmetic Probability

1. This indicates that 30 of the 45 books are paperbacks. Of the 30 paperbacks, 25 could be written in Spanish. In this case, the probability of randomly selecting a paperback book written in Spanish is $\frac{25}{45} > \frac{1}{2}$. On the other hand, it is possible that only 5 of the paperback books are written in Spanish. In this case, the probability of randomly selecting a paperback book written in Spanish is $\frac{5}{45} < \frac{1}{2}$; NOT sufficient.
2. This indicates that 15 of the books are written in Spanish. Then, at most 15 of the 45 books on the shelf are paperbacks written in Spanish, and the probability of randomly selecting a paperback book written in Spanish is at most $\frac{15}{45} < \frac{1}{2}$; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

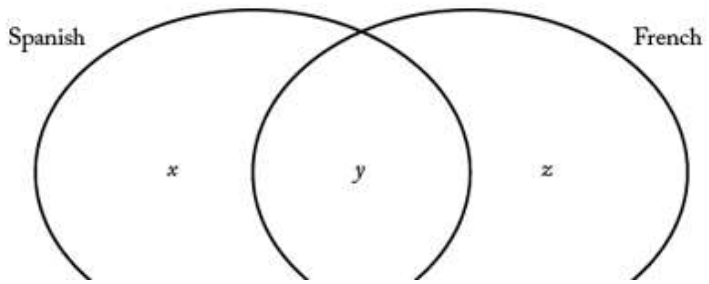
•

A small school has three foreign language classes, one in French, one in Spanish, and one in German. How many of the 34 students enrolled in the Spanish class are also enrolled in the French class?

1. There are 27 students enrolled in the French class, and 49 students enrolled in either the French class, the Spanish class, or both of these classes.
2. One-half of the students enrolled in the Spanish class are enrolled in more than one foreign language class.

Arithmetic Sets

Given that 34 students are enrolled in the Spanish class, how many students are enrolled in both the Spanish and French classes? In other words, given that $x + y = 34$ in the diagram below, what is the value of y?



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1. It is given that $y + z = 27$ and $x + y + z = 49$. Adding the equations $x + y = 34$ and $y + z = 27$ gives $x + 2y + z = 34 + 27 = 61$, or $y + (x + y + z) = 61$. Since $x + y + z = 49$, it follows that $y + 49 = 61$, or $y = 12$; SUFFICIENT.
2. Given that half the students enrolled in the Spanish class are enrolled in more than one foreign language class, then it is possible that no students are enrolled in the French and German classes only and 17 students are enrolled in both the Spanish and French classes. On the other hand, it is also possible that there are 17 students enrolled in the French and German classes only and no students enrolled in both the Spanish and French classes; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

•

Last year $\frac{3}{5}$ of the members of a certain club were males. This year the members of the club include all the members from last year plus some new members. Is the fraction of the members of the club who are males greater this year than last year?

1. More than half of the new members are male.
2. The number of members of the club this year is $\frac{6}{5}$ the number of members last year.

Arithmetic Operations with Fractions

Let L represent the number of members last year; N the number of new members added this year; and x the number of members added this year who are males. It is given that $\frac{3}{5}$ of the members last year were males. It follows that the number of members who are male this year is $\frac{3}{5}L + x$. Also, the total number of members this year is $L + N$.

Determine if $\frac{\frac{3}{5}L + x}{L + N} > \frac{3}{5}$, or equivalently, determine if $3L + 5x > 3L + 3N$ or simply if $x > \frac{3}{5}N$.

1. This indicates that $x > \frac{1}{2}N$. If, for example, $N = 20$ and $x = 11$, then $11 > \frac{1}{2}(20) = 10$, but $11 \not> \frac{3}{5}(20) = 12$. On the other hand, if $N = 20$ and $x = 16$, then $16 > \frac{1}{2}(20) = 10$, and $16 > \frac{3}{5}(20) = 12$; NOT sufficient.
2. This indicates that $L + N = \frac{6}{5}L$. It follows that $N = \frac{1}{5}L$. If, for example, $L = 100$, then $N = \frac{1}{5}(100) = 20$. If $x = 11$, then $11 \not> \frac{3}{5}(20) = 12$. On the other hand, if $x = 16$, then $16 > \frac{1}{2}(20) = 10$, and $16 > \frac{3}{5}(20) = 12$; NOT sufficient.

Taking (1) and (2) together is of no more help than (1) and (2) taken separately since the same examples were used to show that neither (1) nor (2) is sufficient.

The correct answer is E; both statements together are still not sufficient.

•

Machines K, M, and N, each working alone at its constant rate, produce 1 widget in x, y, and 2 minutes, respectively. If Machines K, M, and N work simultaneously at their respective constant rates, does it take them less than 1 hour to produce a total of 50 widgets?

1. $x < 1.5$

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2. $y < 1.2$

Algebra Work Problems

Because Machine N produces 1 widget every 2 minutes, Machine N produces $\frac{60}{2} = 30$ widgets in 1 hour = 60 minutes.

1. Given that $x < 1.5$, it follows that Machine K, which produces $\frac{60}{x} = \text{widgets}$ in 60 minutes, produces more than $\frac{60}{1.5} = 40$ widgets in 1 hour = 60 minutes. Thus, regardless of the number of widgets produced by Machine M, when all three machines are working simultaneously at their respective constant rates, more than 30 + 40 = 70 widgets will be produced in 1 hour. Therefore, the three machines will together have produced 50 widgets in less than 1 hour; SUFFICIENT.
2. Given that $y < 1.2$, it follows that Machine M, which produces $\frac{60}{y} = \text{widgets}$ in 60 minutes, produces more than $\frac{60}{1.2} = 50$ widgets in 1 hour = 60 minutes. Thus, regardless of the number of widgets produced by Machine K, when all three machines are working simultaneously at their respective constant rates, more than 30 + 50 = 80 widgets will be produced in 1 hour. Therefore, the three machines will together have produced 50 widgets in less than 1 hour; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

•

Stations X and Y are connected by two separate, straight, parallel rail lines that are 250 miles long. Train P and train Q simultaneously left Station X and Station Y, respectively, and each train traveled to the other's point of departure. The two trains passed each other after traveling for 2 hours. When the two trains passed, which train was nearer to its destination?

1. At the time when the two trains passed, train P had averaged a speed of 70 miles per hour.
2. Train Q averaged a speed of 55 miles per hour for the entire trip.

Arithmetic Applied Problems; Rates

1. This indicates that Train P had traveled 2(70) = 140 miles when it passed Train Q. It follows that Train P was 250 − 140 = 110 miles from its destination and Train Q was 140 miles from its destination, which means that Train P was nearer to its destination when the trains passed each other; SUFFICIENT.
2. This indicates that Train Q averaged a speed of 55 miles per hour for the entire trip, but no information is given about the speed of Train P. If Train Q traveled for 2 hours at an average speed of 55 miles per hour and Train P traveled for 2 hours at an average speed of 70 miles per hour, then Train P was nearer to its destination when the trains passed. However, if Train Q traveled for 2 hours at an average speed of 65 miles per hour and Train P traveled for 2 hours at an average speed of 60 miles per hour, then Train Q was nearer to its destination when the trains passed. Note that if Train Q traveled at $\frac{(120)(55)}{140} = 47\frac{1}{7}$ miles per hour for the remainder of the trip, then its average speed for the whole trip was 55 miles per hour; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

•

In a two-story apartment complex, each apartment on the upper floor rents for 75 percent as much as each apartment on the lower floor. If the total monthly rent is $15,300 when rent is collected on all of the apartments, what is the monthly rent on each apartment on the lower floor?

1. An apartment on the lower floor rents for $150 more per month than an apartment on the upper floor.
2. There are 6 more apartments on the upper floor than on the lower floor.

Algebra Simultaneous Equations

Let x be the number of apartments on the lower floor, y be the number of apartments on the upper floor, and $\$R$ be the monthly rent on each apartment on the lower floor. Then the monthly rent on each apartment on the upper floor is

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the monthly rent on each apartment on the lower floor. Then the monthly rent on each apartment on the upper floor is $\$0.75R$ and $Rx + 0.75Ry = 15{,}300$. Determine the value of R.

1. Given that $R = 150 + 0.75R$, it follows that $0.25R = 150$, or $R = 600$; SUFFICIENT.
2. Given that $y = x + 6$ thus, $Rx + 0.75R(x + 6) = 15{,}300$, or $1.75Rx + 4.5R = 15{,}300$, which can be true for more than one value of R and a corresponding positive integer value of x. For example, it is possible that $R = 600$ and $x = 12$, and it is possible that $R = 425$ and $x = 18$; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

•

A motorboat, which is set to travel at k kilometers per hour in still water, travels directly up and down the center of a straight river so that the change in the boat's speed relative to the shore depends only on the speed and direction of the current. What is the value of k?

1. It takes the same amount of time for the boat to travel 4 kilometers directly downstream as it takes for it to travel 3 kilometers directly upstream.
2. The current flows directly downstream at a constant rate of 2.5 kilometers per hour.

Algebra Applied Problems

Letting c represent the speed of the current, the boat travels $(k + c)$ kilometers per hour (kph) when traveling in the same direction as the current (downstream) and $(k - c)$ kph when traveling in the opposite direction as the current. Determine the value of k.

1. Given that it takes the same amount of time to travel 4 kilometers (km) downstream as it takes to travel 3 km upstream, it follows that $\frac{4}{k+c} = \frac{3}{k+c}$, or $k = 7c$, which shows that the value of k depends on the value of c; NOT sufficient.
2. Given that $c = 2.5$ kph, it is not possible to determine the value of k since no information is given about the value of k or its relationship with c; NOT sufficient.

Taking (1) and (2) together, $k = 7c = 7(2.5) = 17.5$.

The correct answer is C; both statements together are sufficient.

•

If the book value of a certain piece of equipment was $5,000 exactly 5 years ago, what is its present book value?

1. From the time the piece of equipment was purchased, its book value decreased by 10 percent of its purchase price each year of its life.
2. The present book value of another piece of equipment is $2,000.

Algebra Applied Problems

Determine the present book value of a piece of equipment, of which the book value exactly 5 years ago was $5,000.

1. Given that the book value decreased 10% of the purchase price each year, it is not possible to determine the present book value because the purchase price is unknown as is the number of years ago the equipment was purchased. Consider the following chart, where P represents the purchase price.

Years after purchase	Depreciation	Book value
0	0	P
1	$0.1P$	$0.9P$
2	$0.1P$	$0.8P$
3	$0.1P$	$0.7P$

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4	$0.1P$	$0.6P$
5	$0.1P$	$0.5P$
6	$0.1P$	$0.4P$
7	$0.1P$	$0.3P$
8	$0.1P$	$0.2P$
9	$0.1P$	$0.1P$
10	$0.1P$	0
11	0	0
12	0	0
13	0	0

For example, if "5 years ago" was 6 years after purchase and the book value was \$5,000, then $\$5,000 = 0.4P$ and $P = \$12,500$. In this case, the present book value (i.e., 5 years hence) is \$0. On the other hand, if "5 years ago" was 2 years after purchase and the book value was \$5,000, then $\$5,000 = 0.8P$ and $P = \$6,250$. In this case, the present book value (i.e., 5 years hence) is 0.3(\$6,250) = \$1,875; NOT sufficient.

2. Given that the present book value of another piece of equipment is \$2,000 gives no information about the certain piece of equipment under consideration; NOT sufficient.

Taking (1) and (2) together gives no more information than (1) alone since (2) gives information about another piece of equipment, not the piece under consideration.

The correct answer is E; both statements together are not sufficient.

•

The total cost to charter a bus was shared equally by the people who went on a certain trip. If the total cost to charter the bus was \$360, how many people went on the trip?

1. Each person who went on the trip paid \$9 to charter the bus.
2. If 4 fewer people had gone on the trip, each person's share of the total cost to charter the bus would have increased by \$1.

Algebra First-Degree Equations

Let n be the number of people who went on the trip and $\$p$ be the amount that each person paid. Then $np = 360$. What is the value of n?

1. Given that $p = 9$, then $9n = 360$, or $n = 40$; SUFFICIENT.
2. Given that $(n-4)(p+1) = 360$, or $np + n - 4p - 4 = 360$, it follows from $np = 360$ that $n - 4p - 4 = 0$. Multiplying both sides of this last equation by n gives $n^2 - 4np - 4n = 0$, or $n^2 - 4(360) - 4n = 0$, or $n^2 - 4n - 1{,}440 = 0$. Factoring gives $(n-40)(n+36) = 0$, and hence $n = 40$ since n is a positive integer; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

•

If each of the stamps Carla bought cost 20, 25, or 30 cents and she bought at least one of each denomination, what is the number of 25-cent stamps that she bought?

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1. She spent a total of $1.45 for stamps.
2. She bought exactly 6 stamps.

Arithmetic Applied Problems

Let x, y, and z be the number of 20, 25, and 30 cent stamps, respectively, that Carla bought. What is the value of y?

1. Given that $20x + 25y + 30z = 145$, it is not possible to determine the value of y. For example, if $x = 3$, $y = 1$, and $z = 2$, then each of x, y, and z is a positive integer and $20x + 25y + 30z = 145$. However, if $x = 2$, $y = 3$, and $z = 1$, then each of x, y, and z is a positive integer and $20x + 25y + 30z = 145$; NOT sufficient.
2. Given that $x + y + z = 6$, it is not possible to determine the value of y, because each the two examples used in (1) satisfies $x + y + z = 6$; NOT sufficient.

Taking (1) and (2) together, it is not possible to determine the value of y because the two examples above each satisfy both (1) and (2).

The correct answer is E; both statements together are still not sufficient.

•

A car traveled a distance of d miles in t minutes at an average rate of r miles per minute. What is the ratio of d to r?

1. $t = 30$
2. $d = 25$

Algebra Applied Problems

Determine the ratio of d to r for a car that traveled d miles in t minutes at an average rate of r miles per minute. Note that since $d = rt$ and $\frac{d}{r} = t$, determining the ratio of d to r amounts to determining t.

1. Given that $t = 30$, it follows that $\frac{d}{r} = 30$; SUFFICIENT.
2. Given that $d = 25$, it is not possible to determine the ratio of d to r since no information is given about the value of r nor is information given from which to obtain the value of r; NOT sufficient.

The correct answer is A;statement 1 alone is sufficient.

•

Pat is reading a book that has a total of 15 chapters. Has Pat read at least $\frac{1}{3}$ of the pages in the book?

1. Pat has just finished reading the first 5 chapters.
2. Each of the first 3 chapters has more pages than each of the other 12 chapters in the book.

Arithmetic Applied Problems

Determine if Pat has read at least $\frac{1}{3}$ of the pages in a 15-chapter book.

1. Given that Pat has just finished reading the first 5 chapters in the book, she may or may not have read $\frac{1}{3}$ of the pages in the book. If the first 5 chapters were very short and she has read no other pages in the book, it is possible that she has read less than $\frac{1}{3}$ of the pages in the book. If the first 5 chapters were very long and/or she has read other pages in the book, it is possible that she has read at least $\frac{1}{3}$ of the pages in the book; NOT sufficient.
2. Given that each of the first 3 chapters has more pages than each of the other 12 chapters, it is impossible to determine whether Pat has read at least $\frac{1}{3}$ of the pages in the book because information is not given about how

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many chapters or which chapters of the book Pat has read; NOT sufficient.

Taking (1) and (2) together, if Pat has read just the first 5 chapters and each of the first 3 chapters has 10 pages, each of chapters 4 and 5 has 1 page, and each of the other 10 chapters has 9 pages, then Pat has read 32 pages of the book, which has 32 + 90 = 122 pages and $\frac{32}{122} < \frac{1}{3}$. On the other hand, if Pat has read at least the first 5 chapters, each of the first 3 chapters has 10 pages and each of the other 12 chapters has 9 pages, then Pat has read at least 48 pages of the book, which has 48 + 90 = 138 pages and $\frac{48}{138} > \frac{1}{3}$.

The correct answer is E; both statements together are still not sufficient.

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Should the choices made by 100 participants in a survey about Product X be said to proportionately represent the preferences of the entire population?

1. In the survey of 100 participants, 70% said they preferred Product X.
2. While the survey provided valuable insights into Product X, the small sample may not be representative of the entire population's preferences.

Inference

1. Whatever specific percentage of the survey participants said they preferred Product X, the percentage of individuals with that preference in a much larger population might be exactly the same or much different. Thus, whether the survey participants' choices should be said to proportionately represent the entire population's preferences cannot be determined from (1) alone; NOT sufficient.
2. The statement that a small sample may not be representative of a population's preferences suggests that the sample also may be representative of that population's preferences. Thus, whether the survey participants' choices should be said to proportionately represent the entire population's preferences cannot be determined from (2) alone; NOT sufficient.

If (1) and (2) are both true, it is possible that 70% of the broader population prefers Product X, just as 70% of the survey respondents claimed to prefer it, but it's also possible that some much different percentage of the broader population prefers Product X. Thus, whether the survey participants' choices should be said to proportionately represent the entire population's preferences cannot be determined even from both statements together.

The correct answer is E; both statements together are still not sufficient.

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What is a limiting factor when it comes to us using the most effective solution?

1. We have identified several potential solutions, but a lack of funding hinders their implementation.
2. As the most effective solution requires extensive resources, we must opt for a more affordable option given our budget.

Inference

1. Even if a lack of funding hinders the implementation of several potential solutions we have identified, maybe none of those potential solutions is the most effective solution. Perhaps the most effective solution is well within our budget, but we haven't identified it. Thus, an example of a limiting factor when it comes to us using the most effective solution cannot be determined from (1) alone; NOT sufficient.
2. If the extensive resources required by the most effective solution require us to opt for a more affordable option given our budget, then cost must be one limiting factor when it comes to us using the most effective solution. Thus, (2) alone provides an example of a limiting factor when it comes to us using the most effective solution; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

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What are all the important considerations for making accurate projections?

1. Based on the available data, we believe that the market demand for our product will increase by 20% in the next quarter.

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2. Without comprehensive market research, our projection may not account for external factors that could influence fluctuations in demand.

Inference

1. An observation about how "we" believe market demand for a product will change in the next quarter provides no information about important considerations for making accurate projections. Thus, what considerations are important for making accurate projections cannot be determined from (1) alone; NOT sufficient.
2. (2)'s claim that our projection may not account for external factors without comprehensive market research doesn't tell us what considerations are generally important for making accurate projections. (2) leaves open the possibility that our projection may account for external factors without comprehensive market research. It also leaves open the possibility that the projection may be accurate even if it doesn't account for external factors, or that it may be inaccurate even if it does account for such factors. Furthermore, even if comprehensive market research were one important consideration for making our projection, that wouldn't tell us what the

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